Solution Required Ratio = C.S.A. of the sphere: C.S.A. of the cylinder : C.S.A. of the cone

$= 4\pi r^2 : 2\pi rh : \pi rl \qquad (l = \sqrt{r^2 + h^2} = \sqrt{2r^2} = \sqrt{2}r)$

$= 4\pi r^2 : 2\pi r^2 : \pi r\sqrt{2}\, r$

$= 4\pi r^2 : 2\pi r^2 : \sqrt{2}\pi r^2$

$= 4 : 2 : \sqrt{2} = 2\sqrt{2} : \sqrt{2} : 1$

7.2.7 Frustum of a right circular cone

In olden days a cone shaped buckets [Fig.7.21(a)] filled with sand / water were used to extinguish fire during fire accidents. Later, it was reshaped to a round shaped bottom [Fig.7.21(b)] to increase its volume.



Fig. 7.21(a) Fig. 7.21(b) Fig. 7.21(c)

The shape in [Fig.7.21(c)] resembling a inverted bucket is called as a frustrum of a cone.

The objects which we use in our daily life such as glass, bucket, street cone are examples of frustum of a cone. (Fig.7.22)



Bucket Street Cone Glass

Fig. 7.22

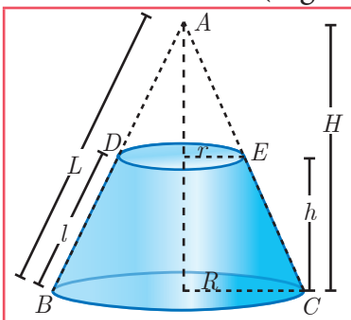


Fig. 7.23

Definition

When a cone ABC is cut through by a plane parallel to its base, the portion of the cone DECB between the cutting plane and the base is called a frustum of the cone.

Surface area of a frustum

Let R and r be radii of the base and top region of the frustum $DECB$ respectively, h is the height and l is the slant height of the same.

Therefore, C.S.A. $= \frac{1}{2}$ (sum of the perimeters of base and top region) × slant height

$= \frac{1}{2}(2\pi R + 2\pi r)l$

C.S.A. of a frustum $= \pi(R + r)l$ sq. units where, $l = \sqrt{h^2 + (R - r)^2}$

T.S.A. = C.S.A. + Area of the bottom circular region
+ Area of the top circular region.

T.S.A. of a frustum $= \pi(R + r)l + \pi R^2 + \pi r^2$ sq. units where, $l = \sqrt{h^2 + (R - r)^2}$

10th_Maths_Chapter 7_English.indd 280 12/3/2024 2:59:36 PM